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=====
TOPOLOGY A – Linear Chain with Bottleneck
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```
h1(10.0.0.1) h3(10.0.0.3) --> s1
s1 --[10 Mbps bottleneck]--> s2
s2 --> h2(10.0.0.2) h4(10.0.0.4)
=====
```

```
SUGGESTED TESTS:
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```
Same-switch:  h1 <--> h3  (expect ~100 Mbps)
Cross-switch: h1 <--> h2  (expect ~10 Mbps)
pingall to check connectivity first
```

```
*** Starting CLI:
```

```
mininet> pingall
```

```
*** Ping: testing ping reachability
```

```
h1 -> h2 h3 h4
```

```
h2 -> h1 h3 h4
```

```
h3 -> h1 h2 h4
```

```
h4 -> h1 h2 h3
```

```
*** Results: 0% dropped (12/12 received)
```

```
mininet> █
```

```
mininet> h3 iperf -s &
```

```
-----
Server listening on TCP port 5001
TCP window size: 85.3 KByte (default)
-----
```

```
mininet> h1 iperf -c 10.0.0.3 -t 10
```

```
-----
Client connecting to 10.0.0.3, TCP port 5001
TCP window size: 85.3 KByte (default)
-----
```

```
[ 1] local 10.0.0.1 port 42572 connected with 10.0.0.3 port 5001 (icwnd/mss/irrtt=14/1448/13874)
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-11.8788 sec  101 MBytes  71.6 Mbits/sec
```

```
mininet> █
```

```
mininet> h2 iperf -s &
```

```
mininet> h1 iperf -c 10.0.0.2 -t 10
```

```
-----
Client connecting to 10.0.0.2, TCP port 5001
TCP window size: 85.3 KByte (default)
-----
```

```
[ 1] local 10.0.0.1 port 53610 connected with 10.0.0.2 port 5001 (icwnd/mss/irrtt=14/1448/59083)
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-13.5424 sec  15.3 MBytes  9.45 Mbits/sec
```

```
mininet> █
```

```
[ 1] 0.000 1575.21 sec  1575 bytes  9.46 bytes/sec
mininet> h1 ping -c 5 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=17.3 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=11.7 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=10.9 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=9.78 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=9.46 ms

--- 10.0.0.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 9.460/11.827/17.259/2.834 ms
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=53.9 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=53.6 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=51.7 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=51.9 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=50.8 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4010ms
rtt min/avg/max/mdev = 50.786/52.384/53.894/1.189 ms
mininet>
```

```
mininet> sh ovs-ofctl dump-ports s1
OFPST_PORT reply (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
               tx pkts=0, bytes=0, drop=0, errs=0, coll=0
  port "s1-eth1": rx pkts=27550, bytes=174047572, drop=0, errs=0, frame=0, over=0, crc=0
                  tx pkts=24222, bytes=1602434, drop=0, errs=0, coll=0
  port "s1-eth2": rx pkts=16033, bytes=1058542, drop=0, errs=0, frame=0, over=0, crc=0
                  tx pkts=18427, bytes=148544808, drop=0, errs=0, coll=0
  port "s1-eth3": rx pkts=8175, bytes=541589, drop=0, errs=0, frame=0, over=0, crc=0
                  tx pkts=9283, bytes=25518417, drop=0, errs=0, coll=0
mininet> sh ovs-ofctl dump-ports s2
OFPST_PORT reply (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
               tx pkts=0, bytes=0, drop=0, errs=0, coll=0
  port "s2-eth1": rx pkts=8126, bytes=536680, drop=0, errs=0, frame=0, over=0, crc=0
                  tx pkts=9327, bytes=25522944, drop=0, errs=0, coll=0
  port "s2-eth2": rx pkts=26, bytes=1916, drop=0, errs=0, frame=0, over=0, crc=0
                  tx pkts=113, bytes=10900, drop=0, errs=0, coll=0
  port "s2-eth3": rx pkts=9283, bytes=25518417, drop=0, errs=0, frame=0, over=0, crc=0
                  tx pkts=8176, bytes=541659, drop=0, errs=0, coll=0
mininet>
```